

CALC 400

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Repository

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Week 7: Exam :: 2

TMV: PS 1

Problem 7.1: Find Derivative of the following function

$$y = \cot^{-1}(4\log_5(x) + \sin^7(3^x))$$

Solution

$$\frac{dy}{dx} = -\frac{1}{1 + (4\log_5(x) + \sin^7(3^x))} * \frac{28(\sin(3^x))^6 \cos(3^x) 3^x \ln(3)}{x \ln(5)}$$

Problem Set: 2

Problem 7.2: Implicit Differentiation

Find $\frac{dx}{dy}$ of the following equation

$$3x^3 + 8x^2y + 3y^2 = 9$$

Problem 7.3: Find Tangent Line

$$y = \log(x^2 - 4x + 1)$$

Problem Set: 3 (MOM)

Problem 7.4: Find Derivative

$$f(x) = \frac{\sqrt{x} - 3}{\sqrt{x} + 3}$$

Step 7.4.1: Step

Step 7.4.2: Step

Week 8: Sec 4.1 and 4.2

Problem Set: Set 4.1 – Related Rates

Problem 8.1:

Problem 8.2:

Problem 8.3:

Problem 8.4:

Problem 8.5: Shadow Problem

A street light is at the top of a 18 ft tall pole. A woman 6 ft tall walks away from the pole with a speed of 4 ft/sec along a straight path. How fast is the tip of her shadow moving when she is 45 ft from the base of the pole?

Step 8.5.1: Find $\frac{dy}{dt}$

$$\frac{18}{6} = \frac{y}{y-x}$$

$$3(y-x) = y$$

$$3y - 3x = y$$

Step 8.5.2: Solve for y

$$y = 3y - 3x$$

$$-2y = -3x$$

$$y = \frac{3}{2}x$$

Step 8.5.3: Find $\frac{dy}{dt}$

$$\frac{dy}{dt} = \frac{3}{2} \frac{dx}{dt}$$

$$\frac{dx}{dt} = 4$$

106.800046875

Solution

$$\frac{dy}{dt} = 6$$

Problem 8.6: Volume Problem

An inverted pyramid is being filled with water at a constant rate of 35 cubic centimeters per second. The pyramid, at the top, has the shape of a square with sides of length 8 cm, and the height is 12 cm.

Step 8.6.1: Formulas and Definitions

$$V = \frac{b^2 h}{3}$$

$$\frac{dV}{dt} = 25$$

$$h = 4$$

$$b = \frac{2}{7}h$$

$$3V = \left(\frac{2}{7}h\right)^2 h$$

$$3V = \frac{4}{49}h^3$$

$$\frac{3V}{\frac{4}{49}} = h^3$$

$$3V * \frac{49}{4} = h^3$$

$$h^3 = \frac{147V}{4}$$

Step 8.6.2: Find $\frac{dh}{dt}$

$$3h^2 * \frac{dh}{dt} = \frac{147 * 25}{4}$$

$$\frac{dh}{dt} = \frac{147 * 25}{12h^2}$$

Solution

$$\frac{dh}{dt} = \frac{147 * 25}{12 * 16}$$

Problem 8.7: Rate of Change

A circle is inside a square.

The radius of the circle is decreasing at a rate of 5 meters per day and the sides of the square are increasing at a rate of 3 meters per day.

When the radius is 2 meters, and the sides are 17 meters, then how fast is the AREA outside the circle but inside the square changing?

The rate of change of the area enclosed between the circle and the square is

Step 8.7.1: Definitions

$$\frac{dr}{dt} = -5$$

$$\frac{ds}{dt} = 3$$

$$BA = sA - cA$$

$$sA = s^2$$

$$\frac{dsA}{dt} = 6s$$

$$cA = \pi r^2$$

$$\frac{dcA}{dt} = -\pi 10r$$

$$\frac{dBA}{dt} = \frac{dsA}{dt} - \frac{dcA}{dt}$$

$$\frac{dBA}{dt} = 6s - (-\pi 10r)$$

Solution

$$\frac{dBA}{dt} = 6 * 17 - (-\pi 10 * 2)$$

Problem 8.8: Speed

A police car is located 60 feet to the side of a straight road.

A red car is driving along the road in the direction of the police car and is 160 feet up the road from the location of the police car. The police radar reads that the distance between the police car and the red car is decreasing at a rate of 100 feet per second. How fast is the red car actually traveling along the road?

The actual speed (along the road) of the red car is feet / second

Step 8.8.1: Definitions

Triangle: abc

$$a = 60$$

$$b = 160$$

$$c = 170.880075$$

$$\frac{dc}{dt} = 100$$

$$\frac{da}{dt} = 0$$

Step 8.8.2: pythagorean theorem

$$b^2 = c^2 - a^2$$

$$2b * \frac{db}{dt} = 2c \frac{dc}{dt} - 2a \frac{da}{dt}$$

$$320 * \frac{db}{dt} = 341.76015 \frac{dc}{dt} - 120 \frac{da}{dt}$$

Step 8.8.3: Solve

$$320 * \frac{db}{dt} = 341.76015 * 100$$

$$\frac{db}{dt} = \frac{341.76015 * 100}{320}$$

Solution

$$\frac{db}{dt} = 106.800046875$$

Problem Set: 4.2 – Linear Approximations and Differentials**Problem 8.9: Square Root Linear Approximation**

Approximate $\sqrt{25.4}$ Let $f(x) = \sqrt{x}$ Find tangent line to $f(x)$ at $x = 25$

Step 8.9.1: Find $f'(x)$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

Step 8.9.2: Slope Intercept Form

$$f'(25) = \frac{1}{10} (25, 5)$$

$$L(x) = \frac{1}{10}(x - 25) + 5$$

$$L(x) = \frac{1}{10}x + 2.5$$

Solution

$$L(25.4) = \frac{1}{10}x + 2.5$$

$$L(25.4) = 2.54 + 2.5$$

$$5.04$$

Problem 8.10: Sin Linear Approximation

$$x = 0 \quad f(x) = \sin(3x)$$

Step 8.10.1: Differentiate

$$f'(x) = \cos(3x) * 3$$

$$(0, 0)$$

Solution

$$L(x) = 3(x)$$

Problem 8.11: Linear Approximation of Complex Root and Percent Error

$$f(x) = \sqrt[3]{-90x^3 + 144}$$

$$f'(x) = \frac{-90x^2}{\sqrt[3]{(-90x^3 + 144)^2}}$$

$$f'(1) = -6.29960524947$$

Problem 8.12: Differentials

$$\frac{dy}{dx} = \sec^2(3x + 6) * 3$$

Solution

$$\frac{dy}{0.1} = 3\sec^2(3x + 6)$$

Problem 8.13: Delta v. Differential Percent Error

$$y = \sin(4x)$$

If $\Delta x = 0.1$ at $x = 1$ Find Δy

Step 8.13.1: Definitions

$$\Delta y = f(x + \Delta x) - f(x)$$

$$\frac{dy}{dx} = f'(x)$$

Step 8.13.2: Delta

$$\Delta y = f(0 + 0.1) - f(0) = 0.389418342309$$

Step 8.13.3: Differentiatals

$$\frac{dy}{dx} = 4\cos(4x)$$

$$dy = \frac{4\cos(4x)}{10}$$

$$dy = 0.4$$

Step 8.13.4: Percent Error

$$100 * \frac{dy - \Delta y}{\Delta y}$$

Solution

$$2.71729822182$$

Group Assignment:**Problem 8.14: Group 6. Sec 4.1 3**

Balloon rises vertically at rate of 10 ft/sec. A person watches the balloon ascend from a point on the ground 100ft away from the spot below the rising balloon. At what rate (radians/second) is the observer's eye rotating upward to follow the balloon when it is 50ft above the level of the observer's eye

Step 8.14.1: Definitions

$$\frac{dh}{dt} = 10$$

$$\frac{db}{dt} = 0$$

$$b = 100$$

$$h = 50$$

$$\tan(\theta) = \frac{h}{b}$$

Step 8.14.2: Setting Up

$$\theta = \tan^{-1}\left(\frac{h}{b}\right)$$

$$u = \frac{h}{b}$$

$$\frac{du}{dt} = \frac{\frac{dh}{dt}b - h\frac{db}{dt}}{b^2}$$

$$\frac{du}{dt} = \frac{\frac{dh}{dt}b}{b^2}$$

$$\frac{du}{dt} = \frac{\frac{dh}{dt}}{b}$$

$$\frac{du}{dt} = \frac{1}{10}$$

$$u = \frac{50}{100}$$

$$u = \frac{1}{2}$$

$$\frac{d\theta}{dt} = \frac{\frac{du}{dt}}{1 + u^2}$$

Step 8.14.3: Plugging In

$$\frac{d\theta}{dt} = \frac{\frac{1}{10}}{\frac{5}{4}}$$
$$\frac{d\theta}{dt} = \frac{1}{10} * \frac{4}{5}$$

Solution

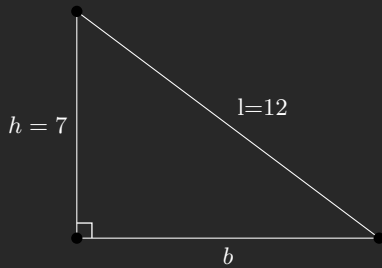
$$\frac{d\theta}{dt} = \frac{4}{50}$$

TMV:

Problem 8.15: # 3

A ladder 12 feet long is leaning against a vertical wall with its other end on the ground. The top end of the ladder is sliding down the wall. When the top end is 7 feet from the ground, it is sliding down the wall at 2 ft/sec. How fast is the bottom moving away from the wall at this instant?

Step 8.15.1: Diagram



Step 8.15.2: Definitions

Sliding Away

$$\frac{dh}{dt} = -2$$

Ladder Length stays the same

$$\frac{dl}{dt} = 0$$

Relationship of Sides

$$b^2 + h^2 = l^2$$

$$b^2 = l^2 - h^2$$

Length of b

$$b = \sqrt{144 - 49}$$

$$b = \sqrt{95}$$

Step 8.15.3: Derivative

$$2b * \frac{db}{dt} = 2l * \frac{dl}{dt} - 2h * \frac{dh}{dt}$$

$$\frac{db}{dt} = \frac{2l * \frac{dl}{dt} - 2h * \frac{dh}{dt}}{2\sqrt{95}}$$

$$\frac{db}{dt} = \frac{0 - (14 * -2)}{2\sqrt{95}}$$

$$\frac{db}{dt} = \frac{14}{\sqrt{95}}$$

Solution

The bottom is sliding at $\frac{14}{\sqrt{95}}$ or 1.43636969 ft/sec

Week 9: Sec 4.3, 4.4 and 4.5

Problem Set: Set 4.3 – Maxima and Minima

Note: Absolute Extrema

Absolute Extrema Consider the function $f(x) = x^2 + 1$ over the interval $(-\infty, \infty)$

Absolute Minima: 1 at $x = 0$ Absolute Maxima: $\nexists$

Most absolute of all local over the interval

Note: Local Extrema

Spots where starts decreasing or increasing in intervals can be multiple per

Absolute is also local

Note: Fermat's Theorem

If f has local extremum at c and f is differentiable at c then $f'(c) = 0$ Note this theorem does not claim that a function f must have a local extremum at a critical point.

Note: Critical Point

$f'(c) = 0$ or $f'(c) = \text{undefined}$

Problem 9.1:

Problem Set: Set 4.4 – The Mean Value Theorem

Note: Rolle's Theorem

Special case of the Mean Value Theorem when f is continuous and derivative over the interval (a, b) and $f(a) = f(b)$ it must have a critical point

Note: Mean Value Theorem

f is continuous over the $[a, b]$ closed interval, and differentiable over the (a, b) open interval then there is a point between a and b where the tangent line at $f(c)$ is parallel to the secant line

Slope of Secant Line is $\frac{f(b)-f(a)}{b-a}$

Applies to Functions with Derivative of Zero $f'(x) = 0$ for all x then $f(x) = \text{constant}$

Problem Set: Set 4.5 – Derivatives and the Shape of a Graph

Note: First Derivative Test

$f'(x) > 0$ increasing $f'(x) < 0$ decreasing

when f' swaps signs then its local maxima or minima

Note: Concavity

if f' is increasing then concave up f' is decreasing then concave down

$f''(x) > 0$ concave up $f''(x) < 0$ concave down

Note: Point of InflectionChanges concavity $f''(x) = 0$ **Note: Second Derivative Test** $f'(c) = 0$ Check to be local minima or maxima $f''(c) > 0$ $c =$ local minima $f''(c) < 0$ $c =$ local maxima**Group Assignment:****Problem 9.2: # 4**Find all critical values and critical points of the function given, then find the regions where the function is decreasing: $f(x) = -x^4 + 8x^2$ **Step 9.2.1: Derivative**

$$f'(x) = -4x^3 + 16x$$

Step 9.2.2: Factor

$$\begin{aligned}
 & -4x^3 + 16x \\
 &= -4x(x^2 - 4) \\
 &= -4x(x - 2)(x + 2)
 \end{aligned}$$

Roots of $f'(x) = x = [-2, 0, 2]$ **Step 9.2.3: Second Derivative**

$$f''(x) = -12x^2 + 16$$

Step 9.2.4: Second Derivative Test

Local Minima

$$f''(0) = 16$$

Local Maxima

$$f''(-2) = -32$$

$$f''(2) = -32$$

Step 9.2.5: Regions

A

$$(-\infty, -2)$$

B

$$(-2, 0)$$

C

$$(0, 2)$$

D

$$(2, \infty)$$

Step 9.2.6: First Derivative Test

A is Increasing

$$f'(-3) = 60$$

B is Decreasing

$$f'(-1) = -12$$

C is Increasing

$$f'(1) = 12$$

D is Decreasing

$$f'(3) = -60$$

Solution

TMV:

Problem 9.3: # 2

Find all critical values and critical points of the function given, then find the regions where the function is decreasing. Find all critical values and critical points using the derivative, and then show your critical values and critical points are true by using the Desmos. Make sure to graph the function and the points. $f'(x) = -2^{\cos(x)}$ on the domain of $[0, 3\pi]$

Step 9.3.1: Derivative

$$f'(x) = -2^{\cos(x)} * \ln(2) * -\sin(x)$$
$$f'(x) = \sin(x)\ln(2) * 2^{\cos(x)}$$

Step 9.3.2: Zeros

$$\sin(n\pi) = 0$$

Thus... $f'(x) = 0$ at $[0, \pi, 2\pi, 3\pi]$

Step 9.3.3: Second Derivative

$$f'(x) = \sin(x)\ln(2) * 2^{\cos(x)}$$
$$f''(x) = \cos(x)\ln(2) * 2^{\cos(x)} + \sin(x)\ln(x) * (-2^{\cos(x)} * \ln(2) * \sin(x))$$

Zeros From Desmos at 1.03366 5.24952 and 7.31685

Step 9.3.4: Intervals

A $(0, \pi)$ is increasing

$$f'(1) = 0.848226443092$$

B $(\pi, 2\pi)$ is decreasing

$$f'(4) = -0.333458413837$$

C $(2\pi, 3\pi)$ is increasing

$$f'(7) = 0.767943315148$$

Step 9.3.5: Local Extremas

π is local maxima

$$f''(\pi) = -0.34657359028$$

2π is local minima

$$f''(2\pi) = 1.38629436112$$

3π is local maxima

$$f''(3\pi) = -0.34657359028$$

Step 9.3.6: Points of Inflection

Set $f''(x) = 0$

1.03366 5.24952 and 7.31685

Solution